

A Comparative Study of Latency-Aware Consensus Protocols

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Abstract

Work in computing research has long relied on assumptions we test directly here. Replicate agreement was high, suggesting the protocol is robust to small handling differences. Residual error grows approximately linearly with the number of coupled parameters. Calibration drift over the trial period remained below the threshold set out in the protocol.

1. Introduction

Longer observation windows narrow the confidence interval but do not shift the central estimate. Neither of the alternative formulations improved on the simple additive model in our setting. Discrepancies at the boundary are consistent with the simplifying assumptions noted above. Replicate agreement was high, suggesting the protocol is robust to small handling differences.

Sensitivity to the smoothing window proved modest across the range we examined. Longer observation windows narrow the confidence interval but do not shift the central estimate. Neither of the alternative formulations improved on the simple additive model in our setting. Residual error grows approximately linearly with the number of coupled parameters.

The measurements reported here were collected over twelve independent trials under controlled conditions. Our baseline reproduces the behaviour described in earlier work to within the stated tolerance. We release the processing scripts and intermediate artefacts to support independent verification. Taken together, these findings argue for treating the two regimes separately in future analyses. Replicate agreement was high, suggesting the protocol is robust to small handling differences. Sensitivity to the smoothing window proved modest across the range we examined. The reduced model retains most of the explanatory power at a fraction of the computational cost. Longer observation windows narrow the confidence interval but do not shift the central estimate. We observe a pronounced inflection once the load exceeds roughly sixty percent of nominal capacity.

2. Related Work

Discrepancies at the boundary are consistent with the simplifying assumptions noted above. Our baseline reproduces the behaviour described in earlier work to within the stated tolerance. The measurements reported here were collected over twelve independent trials under controlled conditions. Sensitivity to the smoothing window proved modest across the range we examined. A modest but repeatable bias appears in the lowest-magnitude bin and warrants further study. Taken together, these findings argue for treating the two regimes separately in future analyses.

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3. Materials and Methods

The measurements reported here were collected over twelve independent trials under controlled conditions. Longer observation windows narrow the confidence interval but do not shift the central estimate. Neither of the alternative formulations improved on the simple additive model in our setting. Calibration drift over the trial period remained below the threshold set out in the protocol. Replicate agreement was high, suggesting the protocol is robust to small handling differences. Residual error grows approximately linearly with the number of coupled parameters. Discrepancies at the boundary are consistent with the simplifying assumptions noted above.

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4. Experimental Setup

Replicate agreement was high, suggesting the protocol is robust to small handling differences. Residual error grows approximately linearly with the number of coupled parameters. Calibration drift over the trial period remained below the threshold set out in the protocol. We release the processing scripts and intermediate artefacts to support independent verification. Discrepancies at the boundary are consistent with the simplifying assumptions noted above.

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5. Results

Neither of the alternative formulations improved on the simple additive model in our setting. Replicate agreement was high, suggesting the protocol is robust to small handling differences. The reduced model retains most of the explanatory power at a fraction of the computational cost. A modest but repeatable bias appears in the lowest-magnitude bin and warrants further study. Longer observation windows narrow the confidence interval but do not shift the central estimate. Taken together, these findings argue for treating the two regimes separately in future analyses.

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6. Discussion

We observe a pronounced inflection once the load exceeds roughly sixty percent of nominal capacity. A modest but repeatable bias appears in the lowest-magnitude bin and warrants further study. Discrepancies at the boundary are consistent with the simplifying assumptions noted above. Calibration drift over the trial period remained below the threshold set out in the protocol. We release the processing scripts and intermediate artefacts to support independent verification. Our baseline reproduces the behaviour described in earlier work to within the stated tolerance. Taken together, these findings argue for treating the two regimes separately in future analyses. Longer observation windows narrow the confidence interval but do not shift the central estimate. The measurements reported here were collected over twelve independent trials under controlled conditions.

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7. Limitations

Longer observation windows narrow the confidence interval but do not shift the central estimate. Discrepancies at the boundary are consistent with the simplifying assumptions noted above. Residual error grows approximately linearly with the number of coupled parameters. We release the processing scripts and intermediate artefacts to support independent verification. Taken together, these findings argue for treating the two regimes separately in future analyses. Sensitivity to the smoothing window proved modest across the range we examined. The measurements reported here were collected over twelve independent trials under controlled conditions. Calibration drift over the trial period remained below the threshold set out in the protocol. A modest but repeatable bias appears in the lowest-magnitude bin and warrants further study.

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